

Assignment 2

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0.1 Instructions

This problem set is based on Lecture 7-11. The question are updated as we cover each lecture.

For each question:

1. Understand the economic intuition before using formulas.
2. Be clear about what the model is trying to answer.
3. Use concise reasoning.

1 Lecture 7: Interest-Rate Models

1.1 Question 1

A junior analyst says:

“Interest rates are uncertain, so we should use a short-rate model for every bond we price.”

The analyst is comparing the following securities:

Security	Description
A	5-year Treasury note with fixed annual coupons
B	5-year fixed-rate corporate bond with no embedded options
C	5-year callable agency bond
D	Interest-rate cap
E	Mortgage-backed security

Answer the following:

- Explain why Securities A and B usually do not require a stochastic short-rate model for time-0 valuation.
- State the basic valuation logic for a plain fixed-cash-flow bond using discount factors.
- Explain what a bootstrapped discount curve provides.
- Identify which securities in the table require future interest-rate dynamics and explain why.
- Correct the analyst’s statement in one or two sentences.

1.2 Question 2

A portfolio manager asks why Lecture 7 spends time modeling the short rate if today’s yield curve is already observable.

Answer the following:

- Explain the difference between a yield curve and a short-rate model.

- b. Explain what is meant by saying that a yield curve is a “snapshot.”
- c. Explain what is meant by saying that a short-rate model is a “scenario generator.”
- d. Explain why future discount curves $P(t, T)$ matter for securities whose cash flows depend on future rates.
- e. Give two examples of portfolio or risk-management questions that cannot be answered well using only today’s yield curve.

1.3 Question 3

Consider the basic one-factor short-rate model:

$$dr_t = b(r_t, t)dt + \sigma(r_t, t)dz_t.$$

Answer the following:

- a. Explain the role of the drift term.
- b. Explain the role of the volatility term.
- c. Explain what mean reversion means and why it may be a reasonable assumption for interest rates.
- d. Compare the intuition behind constant volatility, level-proportional volatility, and square-root volatility.
- e. Explain one practical consequence of choosing a normal model that allows negative rates.

1.4 Question 4

A fixed-income team is deciding between an equilibrium model and an arbitrage-free model.

Answer the following:

- a. Explain the main purpose of an equilibrium interest-rate model.
- b. Explain the main purpose of an arbitrage-free interest-rate model.

- c. Explain why arbitrage-free models are usually preferred for pricing derivatives and bonds with embedded options.
- d. In the Hull-White model,

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t,$$

explain why $\theta(t)$ is central to fitting today's curve.

- e. Explain why a and σ are usually calibrated to interest-rate option prices rather than plain bond prices.

1.5 Question 5

An analyst wants to estimate historical short-rate volatility from weekly data. The analyst observes the following weekly changes in the short rate:

Week	Change in short rate
1	+4 bps
2	-2 bps
3	+6 bps
4	-5 bps
5	+3 bps
6	+1 bp
7	-4 bps
8	+5 bps

Answer the following:

- a. Explain why rate changes, rather than rate levels, are used to estimate volatility in this setting.
- b. Describe the steps for estimating weekly historical volatility from these observations.
- c. Compute the sample standard deviation of the weekly changes in basis points.
- d. Annualize the weekly volatility using $\sqrt{52}$.
- e. Explain two limitations of using this historical volatility estimate as an input to a short-rate model.

2 Lecture 8: Bonds with Embedded Options and Option-Adjusted Measures

2.1 Question 6

A 3-year annual-pay corporate bond has face value 100 and coupon rate 5%. The Treasury zero rates are:

Maturity	Zero rate
1 year	3.00%
2 years	4.00%
3 years	5.00%

The bond trades at 99.00.

Answer the following:

- Write the promised cash flows of the bond.
- Compute the value of the bond if its promised cash flows were discounted using only the Treasury zero curve.
- Explain why the market price can be lower than this Treasury-discounted value.
- Set up the static spread equation for this bond.
- Use the following trial present values to estimate the static spread and explain why the price-spread relationship is nonlinear:

Spread	Present value
0 bps	100.18
50 bps	98.83
100 bps	97.50

2.2 Question 7

A 3-year annual-pay bond has face value 100 and coupon 7. The risk-neutral probability is $p = 0.5$. The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	5%	4%	3%
Middle state		8%	6%
High-rate state			10%

The bond is callable at 101 beginning in year 2.

Answer the following:

- Compute the terminal payoff at year 3.
- Compute the year-2 option-free hold values at the 3%, 6%, and 10% nodes.
- Apply the call rule at year 2. Which nodes are called?
- Roll the callable values back to year 1.
- Compute the callable bond value at time 0 and compare it with the option-free value.

2.3 Question 8

A 3-year annual-pay bond has face value 100 and coupon 4. The risk-neutral probability is $p = 0.5$. The short-rate lattice is:

State	Year 0	Year 1	Year 2
Low-rate state	5%	3%	4%
Middle state		9%	8%
High-rate state			12%

The bond is puttable at 100 beginning in year 1.

Answer the following:

- Compute the terminal payoff at year 3.
- Compute the year-2 option-free hold values.
- Apply the put rule at year 2. Which nodes are put?
- Roll the puttable values back to year 1 and apply the put rule again.
- Compute the puttable bond value at time 0 and compare it with the option-free value.

2.4 Question 9

An analyst says:

“A binomial interest-rate tree is determined by the starting short rate, volatility, and time step, so it cannot also fit the full zero curve.”

Assume a model uses the following structure:

$$r_{t,j} = a_t e^{(2j-t)\sigma}.$$

Answer the following:

- Explain what σ controls in this tree.
- Explain what the time-dependent level parameter a_t controls.
- Explain why a valuation-grade tree needs one calibration condition for each maturity on the zero curve.
- Suppose the zero curve has four annual maturities. How many level parameters are needed to fit the four discount factors?
- Correct the analyst’s statement in one or two sentences.

2.5 Question 10

A callable bond is priced with a calibrated 4-period lattice. The model price with OAS equal to 0 is 103.25, while the observed market price is 101.00. The analyst then tries several OAS values:

Trial OAS	Model price
0 bps	103.25
50 bps	101.80
75 bps	101.08
80 bps	100.94

After solving for OAS, the analyst computes effective duration and convexity using a 50 bp parallel curve shock. The model produces:

Scenario	Repriced value
Base curve	101.00
Curve down 50 bps	103.20
Curve up 50 bps	98.90

Answer the following:

- a. Explain what OAS is adding to the lattice.
- b. Estimate the OAS from the trial values.
- c. Explain why the OAS is positive in this example.
- d. Compute effective duration.
- e. Compute effective convexity and explain what these measures capture for a callable bond.